

Third-Party Integrations

Payments, identity, messaging, maps, Google and Azure dependencies.



This document explains its subject first in plain language, then in engineering detail. It is grounded in the Craves repository snapshot and deliberately separates current implementation, legacy/reference code, milestone foundations and repository gaps so readers do not mistake aspiration for proof.

Version 1.0 | Evidence date: 14 August 2026 | Repository: [rmorampudi09-arch/Craves-Build-platform](#) | Snapshot commit: [3225f7fa531a6b07185fb5e034f7930f8bd8b571](#)

provider adapter pattern - plain-language purpose

Plain-English explanation

Craves includes a governed semantic meal concierge for assisted discovery rather than an authority that can bypass commerce rules.

How it works technically

Recent main history and [apps/customer-web-next/docs/signalr-ai-concierge.md](#) describe SignalR semantic concierge work. Recommendations still rely on governed product data and ordinary catalog, quote, authorization and order paths for transactions.

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1. Actor starts the capability with the appropriate identity/context. 2. Client/BFF validates local prerequisites and sends an API request. 3. APIM and the owning service enforce identity, role, ownership and business-state rules. 4. The domain service reads/writes authoritative state and calls providers only through defined boundaries. 5. The caller receives authoritative result or a traceable error; support uses request/correlation evidence rather than secrets.

Errors, edge cases and controls

- Stale UI/cache must not override server state.
- UI visibility is not authorization evidence.
- Retries of money/state-changing operations require idempotency awareness.
- Provider/configuration outages should preserve core domain consistency where safe.
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Diagnostic and validation steps

1. Capture time, environment, operation, expected/actual result and a user-safe identifier. 2. Capture HTTP status, stable error code and request/correlation ID; never copy OTPs, passwords, access tokens or provider secrets. 3. Verify caller role/ownership and current authoritative state before retrying. 4. Check owning service health, then its provider/dependency after local/configuration causes are excluded. 5. If a release is implicated, compare artifact/config marker with prior known-good revision and validate rollback compatibility.

Data, security and deployment impact

The ai boundary owns its authoritative state; clients request or display changes but do not become the source of truth. Secrets and unnecessary personal data stay outside logs and PDFs. Releases must keep API, policy, schema and artifact compatible, with health/readiness evidence and rollback/backout appropriate to the change.

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integration-service is the provider boundary so third-party APIs do not leak provider-specific rules into every domain.

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Cashfree - plain-language purpose

Plain-English explanation

Payments are separated from raw card handling so the Craves UI can use provider-hosted entry while the backend protects order/payment state.

How it works technically

The web documents Cashfree hosted payment sessions. order-service verifies Razorpay callbacks with HMAC SHA-256 over the raw body using X-Razorpay-Signature, and integration-service documents normalized payment-link operations.

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Razorpay webhook HMAC - plain-language purpose

Plain-English explanation

Payments are separated from raw card handling so the Craves UI can use provider-hosted entry while the backend protects order/payment state.

How it works technically

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1. Capture time, environment, operation, expected/actual result and a user-safe identifier. 2. Capture HTTP status, stable error code and request/correlation ID; never copy OTPs, passwords, access tokens or provider secrets. 3. Verify caller role/ownership and current authoritative state before retrying. 4. Check owning service health, then its provider/dependency after local/configuration causes are excluded. 5. If a release is implicated, compare artifact/config marker with prior known-good revision and validate rollback compatibility.

Data, security and deployment impact

The payment boundary owns its authoritative state; clients request or display changes but do not become the source of truth. Secrets and unnecessary personal data stay outside logs and PDFs. Releases must keep API, policy, schema and artifact compatible, with health/readiness evidence and rollback/backout appropriate to the change.

Evidence status

Implemented on main. Evidence: apps/customer-web-next/README.md; apps/customer-web-next/src/lib/cashfree/; services/order-service/README.md; services/integration-service/README.md. Snapshot: main @ 3225f7fa531a (14 August 2026).

refund/provider state - deployment, security and operational validation

Plain-English explanation

Payments are separated from raw card handling so the Craves UI can use provider-hosted entry while the backend protects order/payment state.

How it works technically

The web documents Cashfree hosted payment sessions. order-service verifies Razorpay callbacks with HMAC SHA-256 over the raw body using X-Razorpay-Signature, and integration-service documents normalized payment-link operations.

Flow described in words

1. Actor starts the capability with the appropriate identity/context. 2. Client/BFF validates local prerequisites and sends an API request. 3. APIM and the owning service enforce identity, role, ownership and business-state rules. 4. The domain service reads/writes authoritative state and calls providers only through defined boundaries. 5. The caller receives authoritative result or a traceable error; support uses request/correlation evidence rather than secrets.

Errors, edge cases and controls

- Stale UI/cache must not override server state.
- UI visibility is not authorization evidence.
- Retries of money/state-changing operations require idempotency awareness.
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Azure Communication Services - plain-language purpose

Plain-English explanation

Craves includes a governed semantic meal concierge for assisted discovery rather than an authority that can bypass commerce rules.

How it works technically

Recent main history and apps/customer-web-next/docs/signalr-ai-concierge.md describe SignalR semantic concierge work. Recommendations still rely on governed product data and ordinary catalog, quote, authorization and order paths for transactions.

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Evidence status

Implemented on main. Evidence: apps/customer-web-next/docs/signalr-ai-concierge.md; commit 71e795b. Snapshot: main @ 3225f7fa531a (14 August 2026).

Azure Communication Services - technical architecture

Plain-English explanation

The active backend target is seven Spring Boot services: auth, user-chef, catalog, order, subscription, integration and notification.

How it works technically

Services own focused business domains, validate JWT role/ownership, expose health endpoints, evolve schema through Flyway where documented, and isolate provider-specific behavior behind integration/notification boundaries.

Flow described in words

1. Actor starts the capability with the appropriate identity/context. 2. Client/BFF validates local prerequisites and sends an API request. 3. APIM and the owning service enforce identity, role, ownership and business-state rules. 4. The domain service reads/writes authoritative state and calls providers only through defined boundaries. 5. The caller receives authoritative result or a traceable error; support uses request/correlation evidence rather than secrets.

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Data, security and deployment impact

The backend boundary owns its authoritative state; clients request or display changes but do not become the source of truth. Secrets and unnecessary personal data stay outside logs and PDFs. Releases must keep API, policy, schema and artifact compatible, with health/readiness evidence and rollback/backout appropriate to the change.

Evidence status

Implemented on main. Evidence: services/auth-service/README.md; services/user-chef-service/README.md; services/catalog-service/README.md; services/order-service/README.md; services/subscription-service/README.md; services/integration-service/README.md; services/notification-service/README.md. Snapshot: main @ 3225f7fa531a (14 August 2026).

Azure Communication Services - end-to-end flow

Plain-English explanation

The active backend target is seven Spring Boot services: auth, user-chef, catalog, order, subscription, integration and notification.

How it works technically

Services own focused business domains, validate JWT role/ownership, expose health endpoints, evolve schema through Flyway where documented, and isolate provider-specific behavior behind integration/notification boundaries.

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Azure Communication Services - errors and edge cases

Plain-English explanation

A useful failure model separates user input/state conflicts, identity/permission failures, dependency outages, data faults and deployment/configuration incidents.

How it works technically

order-service documents Problem Details fields and codes including ORDER_NOT_FOUND, ORDER_IDEMPOTENCY_CONFLICT, INVALID_STATUS, CANCELLATION_NOT_ALLOWED, REFUND_FAILED, UNAUTHORIZED, FORBIDDEN, BACKEND_CHANGE_DISABLED and BACKEND_UNAVAILABLE. Diagnostics should start from request/correlation evidence.

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Evidence status

Implemented on main. Evidence: services/order-service/README.md; services/notification-service/README.md; apps/customer-web-next/README.md. Snapshot: main @ 3225f7fa531a (14 August 2026).

Azure Communication Services - deployment, security and operational validation

Plain-English explanation

The repository has broad CI/CD references across backend, frontend, mobile, APIM, data, privacy, security, smoke and release controls.

How it works technically

customer-web-next README describes ACR build/push, Container App revision deployment, readiness checks and restoration of the prior image/revision on failed readiness. Pipeline names not resolvable at their cited path are classified as reference-only instead of assigned invented triggers.

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Data, security and deployment impact

The devops boundary owns its authoritative state; clients request or display changes but do not become the source of truth. Secrets and unnecessary personal data stay outside logs and PDFs. Releases must keep API, policy, schema and artifact compatible, with health/readiness evidence and rollback/backout appropriate to the change.

Evidence status

Implemented on main. Evidence: apps/customer-web-next/README.md; repository YAML search; docs/handover/. Snapshot: main @ 3225f7fa531a (14 August 2026).

FCM - plain-language purpose

Plain-English explanation

Craves includes a governed semantic meal concierge for assisted discovery rather than an authority that can bypass commerce rules.

How it works technically

Recent main history and [apps/customer-web-next/docs/signalr-ai-concierge.md](#) describe SignalR semantic concierge work. Recommendations still rely on governed product data and ordinary catalog, quote, authorization and order paths for transactions.

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FCM - technical architecture

Plain-English explanation

integration-service is the provider boundary so third-party APIs do not leak provider-specific rules into every domain.

How it works technically

Repository documentation covers Google Maps fallback, Calendar creation, Gmail drafts, user grants, contact aliasing, pseudonymized audit, Razorpay requirements/payment links and deterministic disabled-by-default stubs.

Flow described in words

1. Actor starts the capability with the appropriate identity/context. 2. Client/BFF validates local prerequisites and sends an API request. 3. APIM and the owning service enforce identity, role, ownership and business-state rules. 4. The domain service reads/writes authoritative state and calls providers only through defined boundaries. 5. The caller receives authoritative result or a traceable error; support uses request/correlation evidence rather than secrets.

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Evidence status

Implemented on main. Evidence: services/integration-service/README.md; services/integration-service/**/README.md. Snapshot: main @ 3225f7fa531a (14 August 2026).

FCM - end-to-end flow

Plain-English explanation

integration-service is the provider boundary so third-party APIs do not leak provider-specific rules into every domain.

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FCM - errors and edge cases

Plain-English explanation

A useful failure model separates user input/state conflicts, identity/permission failures, dependency outages, data faults and deployment/configuration incidents.

How it works technically

order-service documents Problem Details fields and codes including ORDER_NOT_FOUND, ORDER_IDEMPOTENCY_CONFLICT, INVALID_STATUS, CANCELLATION_NOT_ALLOWED, REFUND_FAILED, UNAUTHORIZED, FORBIDDEN, BACKEND_CHANGE_DISABLED and BACKEND_UNAVAILABLE. Diagnostics should start from request/correlation evidence.

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Evidence status

Implemented on main. Evidence: services/order-service/README.md; services/notification-service/README.md; apps/customer-web-next/README.md. Snapshot: main @ 3225f7fa531a (14 August 2026).

FCM - deployment, security and operational validation

Plain-English explanation

The repository has broad CI/CD references across backend, frontend, mobile, APIM, data, privacy, security, smoke and release controls.

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customer-web-next README describes ACR build/push, Container App revision deployment, readiness checks and restoration of the prior image/revision on failed readiness. Pipeline names not resolvable at their cited path are classified as reference-only instead of assigned invented triggers.

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The devops boundary owns its authoritative state; clients request or display changes but do not become the source of truth. Secrets and unnecessary personal data stay outside logs and PDFs. Releases must keep API, policy, schema and artifact compatible, with health/readiness evidence and rollback/backout appropriate to the change.

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Implemented on main. Evidence: apps/customer-web-next/README.md; repository YAML search; docs/handover/. Snapshot: main @ 3225f7fa531a (14 August 2026).

Service Bus - plain-language purpose

Plain-English explanation

Notifications centralize email, SMS and push so domain services do not each implement vendor logic.

How it works technically

notification-service documents Azure Communication Services, FCM, preferences, device tokens, optional Service Bus topic craves-domain-events, event-id idempotency, correlation propagation and a seven-day resend worker.

Flow described in words

1. Actor starts the capability with the appropriate identity/context. 2. Client/BFF validates local prerequisites and sends an API request. 3. APIM and the owning service enforce identity, role, ownership and business-state rules. 4. The domain service reads/writes authoritative state and calls providers only through defined boundaries. 5. The caller receives authoritative result or a traceable error; support uses request/correlation evidence rather than secrets.

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Data, security and deployment impact

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Evidence status

Implemented on main. Evidence: services/notification-service/README.md; apps/customer-web-next/src/components/notifications/. Snapshot: main @ 3225f7fa531a (14 August 2026).

Service Bus - technical architecture

Plain-English explanation

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How it works technically

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Service Bus - end-to-end flow

Plain-English explanation

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Flow described in words

1. Actor starts the capability with the appropriate identity/context. 2. Client/BFF validates local prerequisites and sends an API request. 3. APIM and the owning service enforce identity, role, ownership and business-state rules. 4. The domain service reads/writes authoritative state and calls providers only through defined boundaries. 5. The caller receives authoritative result or a traceable error; support uses request/correlation evidence rather than secrets.

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- Stale UI/cache must not override server state.
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Data, security and deployment impact

The notification boundary owns its authoritative state; clients request or display changes but do not become the source of truth. Secrets and unnecessary personal data stay outside logs and PDFs. Releases must keep API, policy, schema and artifact compatible, with health/readiness evidence and rollback/backout appropriate to the change.

Evidence status

Implemented on main. Evidence: services/notification-service/README.md; apps/customer-web-next/src/components/notifications/. Snapshot: main @ 3225f7fa531a (14 August 2026).

Service Bus - deployment, security and operational validation

Plain-English explanation

Notifications centralize email, SMS and push so domain services do not each implement vendor logic.

How it works technically

notification-service documents Azure Communication Services, FCM, preferences, device tokens, optional Service Bus topic craves-domain-events, event-id idempotency, correlation propagation and a seven-day resend worker.

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Evidence status

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event idempotency - plain-language purpose

Plain-English explanation

Order management is the transactional spine of Craves: create, retrieve, status, cancellation/refund, delivery state, ETA/shortage, proof and administrative controls.

How it works technically

order-service documents idempotency, structured Problem Details errors, reconciliation, provider integration and controlled multi-database routing. Supported roles include customer, chef, admin and delivery-partner contexts with ownership enforcement.

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Evidence status

Implemented on main. Evidence: services/order-service/README.md; apps/customer-web-next/src/components/order-history/; apps/customer-web-next/src/components/order-tracking/. Snapshot: main @ 3225f7fa531a (14 August 2026).

event idempotency - technical architecture

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event idempotency - end-to-end flow

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Google Maps - plain-language purpose

Plain-English explanation

Location/address features help a customer provide a deliverable address while retaining manual fallback when device permission or geocoding fails.

How it works technically

The web contains location/maps/address-dialog modules. Maps are an external dependency; UI convenience does not replace backend serviceability and address validation.

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Evidence status

Implemented on main. Evidence: apps/customer-web-next/src/components/location/; apps/customer-web-next/src/components/maps/; apps/customer-web-next/src/lib/location/; docs/handover/2026-08-06-customer-chef-precise-ui-address-dialog.md. Snapshot: main @ 3225f7fa531a (14 August 2026).

Google Maps - technical architecture

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